

Wigner distribution associated with linear canonical transform of generalized 2-D analytic signals

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ABSTRACT

Analytic complex signals find numerous applications in image and signal processing community. This article focuses on the discussion and definitions of three generalized two-dimensional (2-D) analytic signals (GASs). It begins by explaining the parametric Hilbert transform (PHT) and parametric Riesz transform (PRT), derived from analyzing the irrotational and solenoidal vector fields, respectively. Furthermore, the article explores the four-dimensional Wigner distributions associated with the linear canonical transform (WDLs) for these GASs. Finally, the advantages of these GASs are demonstrated through their application in envelope detection, and highlighting these WDLs potential for feature extraction. This article provides valuable insights into the field of 2-D analytic signals and their applications in various domains.

1. Introduction

Two-dimensional (2-D) analytic signals, also known as complex images, are a generalization of one-dimensional (1-D) analytic signals [1,2] to two dimensions. They play a pivotal role in signal processing and communication, image processing and computer vision, medical imaging and biological signal processing, non-destructive testing and material analysis, as well as audio processing and music analysis. These signals enable precise signal manipulation, feature extraction, diagnostic support, monitoring assistance, material performance evaluation, audio enhancement, and overall advancement in science and technology applications [3–5].

The 2-D analytic signals has been well developed. Currently, there exist multiple definitions of 2-D analytic signals based on different methodologies. In [6], an extension of the concept of analytic signals to multidimensional signals is presented. This extension involves obtaining the real and imaginary parts of the signal through a linear combination of the original signal and its total and partial Hilbert transforms. Bülow [7,8] has introduced a novel definition of analytic signal utilizing the quaternion Fourier transform (QFT). By replacing the complex frequency domain with the quaternion frequency domain defined by the QFT, this approach overcomes the limitation of being unable to reconstruct the original real signal from its complex analytic signal. Felsberg [9] derived the Riesz transform through the analysis of irrotational and

solenoidal vector fields. When applied to a 2-D signal, the Riesz transform generates a 2-D analytic signal in the quaternion domain, referred to as a monogenic signal [10,11]. These 2-D analytic signals have been employed in various fields, such as quaternion wavelet analysis, edge detection, and feature extraction [12,13,5,14]. In addition to discussing the definitions of these 2D analytical signals themselves, it is also essential to explore their application in time-frequency analysis to fully leverage their advantages [15–18].

Numerous research outcomes have been achieved regarding 2-D analytic signals; however, there are limitations in further enhancing processing performance. To address this, an effective solution is to introduce a novel transformation method, such as the linear canonical transform (LCT) [19–21] discussed in this article. The LCT is a three-free-parameter (a, b, c, d) class of integral transform. It is a more versatile tool than the Fourier transform (FT) and plays a more important role in signal processing. Significant research progress has been made in defining two-dimensional analytical signals based on the LCT as presented in [22]. The paper introduces a generalized 2-D analytic signal (GAS) by integrating the total and partial Hilbert transforms associated with the LCT. Furthermore, the proposed approach demonstrates successful applications in envelope detection.

The main objective of this article is to discuss the definitions of three 2-D analytic signals. To establish these definitions, the article begins by explaining the parametric Hilbert transform (PHT) and parametric

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Riesz transform (PRT). Notably, the PHT and PRT are derived through the analysis of the irrotational vector field and the solenoidal vector field, respectively. Moreover, to demonstrate the advantages and significance of these generalized 2-D analytic signals (GASs), we analyze the effects of three types of these signals in envelope detection applications. Furthermore, we delve into the analysis of the four-dimensional Wigner distributions associated with the LCT (4-D WDLs) of these three types of generalized 2-D analytic signals. This is achieved by employing the correlation product of the aforementioned signals. We subsequently explore the potential applications of these WDLs as powerful tools for feature extraction and propose that they could be integrated with machine learning algorithms in future research. In summary, the main contributions of this article can be divided into the following three points: (1) The PHT and the PRT are derived by analyzing irrotational and solenoidal vector fields. (2) Three types of generalized 2-D analytic signals are defined based on the PHT and PRT. (3) Three types of WDLs are defined based on the correlation product of the aforementioned GASs.

The paper is organized as follows. In Section 2, we provide a brief introduction to the basic knowledge of quaternion algebra, 2-D complex signals, 2-D quaternionic signal, 2-D monogenic signal, LCT and quaternion LCT (QLCT). In Section 3, we first derive the 1-D parametric Hilbert transform by means of 2-D harmonic fields. Then the definition of partial PHT (PPHT) and total PHT (TPHT) are given, which in turn defined generalized 2-D complex signals. In Section 4, the generalized 2-D quaternionic signal is defined. In Section 5, we derive the PRT by means of 3-D harmonic fields and define the generalized 2-D monogenic signal. In Section 6, three types of 4-D WDLs and 4-D AFLs are defined based on the correlation product of aforementioned GASs. Next, the potential applications of the three types of generalized 2-D analytical signals and their 4-D WDLs are discussed in Section 7. Finally, we conclude this paper in Section 8.

2. Preliminaries

2.1. The quaternion algebra

The quaternions, one of the hypercomplex numbers, are normed and noncommutative division algebras in which every nonzero element has an inverse. The set of quaternions is indicated by $\mathbb{H}$. Every element of $\mathbb{H}$ can be expressed in the form of Cartesian product [20]

$$\mathbb{H} = \{q \mid q = q_0 + \mathbf{i}q_1 + \mathbf{j}q_2 + \mathbf{k}q_3; q_0, q_1, q_2, [q]_3 \in \mathbb{R}\}, \quad (1)$$

where the units $\mathbf{i}, \mathbf{j}, \mathbf{k}$ obey Hamilton's multiplication rules

$$\mathbf{i}^2 = -\mathbf{j}^2 = -\mathbf{k}^2 = -1; \mathbf{j}\mathbf{k} = -\mathbf{k}\mathbf{j} = \mathbf{i}, \mathbf{k}\mathbf{i} = -\mathbf{i}\mathbf{k} = \mathbf{j}, \mathbf{i}\mathbf{j} = -\mathbf{j}\mathbf{i} = \mathbf{k}. \quad (2)$$

The conjugate of a quaternion q is $q^* = q_0 - \mathbf{i}q_1 - \mathbf{j}q_2 - \mathbf{k}q_3$, and the norm of q is defined as $|q| = \sqrt{qq^*} = \sqrt{q_0^2 + q_1^2 + q_2^2 + q_3^2}$.

In quaternionic notation, let $f: \mathbb{R}^2 \rightarrow \mathbb{H}$ be a quaternion function, then

$$f(\mathbf{t}) = f_0(\mathbf{t}) + \mathbf{i}f_1(\mathbf{t}) + \mathbf{j}f_2(\mathbf{t}) + \mathbf{k}f_3(\mathbf{t}), \quad (3)$$

where $\mathbf{t} = (t_1, t_2)^T \in \mathbb{R}^2$ ($\mathbf{T}$ indicates the transpose). The quaternion module $L^2(\mathbb{R}^2, \mathbb{H})$ is defined as $L^2(\mathbb{R}^2, \mathbb{H}) = \{f \mid f: \mathbb{R}^2 \rightarrow \mathbb{H}, \|f\| = (\int_{\mathbb{R}^2} |f(\mathbf{t})|^2 d\mathbf{t})^{1/2} < \infty\}$, where $d\mathbf{t} = dt_1 dt_2$. Consider two quaternion functions $f, g \in L^2(\mathbb{R}^2, \mathbb{H})$, the inner product of f, g is $(f, g) = \int_{\mathbb{R}^2} f(\mathbf{t})g^*(\mathbf{t})d\mathbf{t}$.

2.2. Two-dimensional complex signals

Consider a 2-D real function $f(t_1, t_2) \in \mathbb{R}^2$. The 2-D FT of f is defined by [19]

$$F(\omega_1, \omega_2) = \iint f(t_1, t_2) e^{-i\omega_1 t_1} e^{-i\omega_2 t_2} dt_1 dt_2. \quad (4)$$

The spectrum obeys the rules of Hermitian symmetry $F(\omega_1, \omega_2) = F^*(-\omega_1, -\omega_2)$ and $F(-\omega_1, \omega_2) = F^*(\omega_1, -\omega_2)$. Therefore, the signal can be reconstructed through its half-plane spectrum based on this symmetry.

A single orthant spectrum can be obtained by the multiplication of $F(\omega_1, \omega_2)$ by the 2-D unit step function as follows [23]

$$\mathbf{1}(\omega_1, \omega_2) = \begin{cases} 1, & \omega_1 > 0, \omega_2 > 0 \\ 1/2, & \omega_1 = 0, \omega_2 = 0 \\ 0, & \omega_1 < 0, \omega_2 < 0 \end{cases}. \quad (5)$$

The 2-D unit step frequency-domain operators in the first and third quadrants of (ω_1, ω_2) are given by

$$\begin{aligned} 4\mathbf{1}(\omega_1, \omega_2) &= [1 + \text{sgn}(\omega_1)] [1 + \text{sgn}(\omega_2)], \\ 4\mathbf{1}(-\omega_1, -\omega_2) &= [1 - \text{sgn}(\omega_1)] [1 - \text{sgn}(\omega_2)]. \end{aligned} \quad (6)$$

The inverse FT yields the 2-D complex analytic signals with single-quadrant spectrum of the form [8]

$$\varphi_1(t_1, t_2) = f - h - \mathbf{i}(h_1 + h_2), \quad \varphi_2(t_1, t_2) = f - h + \mathbf{i}(h_1 + h_2), \quad (7)$$

where $h_1(t_1, t_2) = H_1\{f\} = f(t_1, t_2) * \frac{\delta(t_2)}{-\pi t_1} = -\frac{1}{\pi} \text{p.v.} \int \frac{f(\xi_1, t_2)}{(t_1 - \xi_1)} d\xi_1$, $h_2(t_1, t_2) = H_2\{f\} = f(t_1, t_2) * \frac{\delta(t_1)}{-\pi t_2} = -\frac{1}{\pi} \text{p.v.} \int \frac{f(t_1, \xi_2)}{(t_2 - \xi_2)} d\xi_2$ are the partial Hilbert transforms of f , and $h(t_1, t_2) = H\{f\} = f(t_1, t_2) * \frac{1}{\pi^2 t_1 t_2} = \frac{1}{\pi^2} \text{p.v.} \iint \frac{f(\xi_1, \xi_2)}{(t_1 - \xi_1)(t_2 - \xi_2)} d\xi_1 d\xi_2$ is the total Hilbert transform of f , and the symbol $*$ denotes the 2-D convolution.

2.3. Two-dimensional quaternionic signals

Consider a 2-D real signal $f(t_1, t_2)$, the quaternionic Fourier transform (QFT) of it is defined by the integral [24]

$$F_Q(\omega_1, \omega_2) = \iint e^{-i\omega_1 t_1} f(t_1, t_2) e^{-j\omega_2 t_2} dt_1 dt_2. \quad (8)$$

The QFT is invertible and its inverse is given by the integral

$$f(t_1, t_2) = \iint e^{i\omega_1 t_1} F_Q(\omega_1, \omega_2) e^{j\omega_2 t_2} d\omega_1 d\omega_2. \quad (9)$$

The spectrum in the first quadrant is given by $4\mathbf{1}(\omega_1, \omega_2)F_Q(\omega_1, \omega_2)$. The inverse QFT of this spectrum yields the quaternionic signal with single-quadrant spectrum of the form

$$\begin{aligned} \varphi_Q(\mathbf{t}) &= 4 \int_0^\infty \int_0^\infty e^{i\omega_1 t_1} F_Q(\omega_1, \omega_2) e^{j\omega_2 t_2} d\omega_1 d\omega_2 \\ &= f(\mathbf{t}) - \mathbf{i}h_1(\mathbf{t}) - \mathbf{j}h_2(\mathbf{t}) + \mathbf{k}h(\mathbf{t}), \end{aligned} \quad (10)$$

where $\mathbf{t} = (t_1, t_2)^T$, and h_1, h_2, h are the same as used in (7). The conjugate of $\varphi_Q(\mathbf{t})$ is [25]

$$\varphi_Q^*(\mathbf{t}) = f(\mathbf{t}) + \mathbf{i}h_1(\mathbf{t}) + \mathbf{j}h_2(\mathbf{t}) - \mathbf{k}h(\mathbf{t}). \quad (11)$$

It can be seen that a single quaternionic signal replaces two analytic signals.

2.4. Two-dimensional monogenic signal

The 2-D monogenic signal is derived using Riesz transforms. The Riesz kernels in $\mathbf{t} = (t_1, t_2)^T$ and their 2-D FTs are given by [15]

$$r_1(\mathbf{t}) = -\frac{t_1}{2\pi \left(\sqrt{t_1^2 + t_2^2}\right)^3} \xrightarrow{2D FT} \mathcal{H}_{R_{11}} = \frac{i\omega_1}{\sqrt{\omega_1^2 + \omega_2^2}}, \quad (12)$$

$$r_2(\mathbf{t}) = -\frac{t_2}{2\pi \left(\sqrt{t_1^2 + t_2^2}\right)^3} \xrightarrow{2D FT} \mathcal{H}_{R_{12}} = \frac{j\omega_2}{\sqrt{\omega_1^2 + \omega_2^2}}, \quad (13)$$

where $\mathbf{i}, \mathbf{j}$ is the imaginary part and $\mathcal{H}_{R_{i,k}}$ represents the 2D FT of r_k , with $k = 1, 2$ as the index. The Riesz transforms is defined as a 2-D convolution between the functions f and $r_k, k = 1, 2$.

$$f_{R_{i,k}}(\mathbf{t}) = f(\mathbf{t}) * r_{i,k}(\mathbf{t}). \quad (14)$$

The 2-D monogenic signal has the form [9]

$$\varphi_M(\mathbf{t}) = f - \mathbf{i}f_{R_{i,1}}(\mathbf{t}) - \mathbf{j}f_{R_{j,2}}(\mathbf{t}). \quad (15)$$

2.5. Quaternion linear canonical transform

The 1-D LCT [19,21] of signal f with $A = (a, b; c, d), ad - bc = 1$, and $b \neq 0$ is defined by

$$L_A\{f\}(\omega) = \frac{1}{\sqrt{i2\pi b}} \int f(t) e^{i\left(\frac{a}{2b}t^2 - \frac{1}{b}t\omega + \frac{d}{2b}\omega^2\right)} dt. \quad (16)$$

The 2-D LCT is defined by [21]

$$L_A\{f\}(\omega_1, \omega_2) = \int_{\mathbb{R}^2} f(t_1, t_2) K_{A_1}(t_1, \omega_1) K_{A_2}(t_2, \omega_2) dt, \quad (17)$$

where $A_k = (a_k, b_k; c_k, d_k), a_k d_k - b_k c_k = 1, b_k \neq 0, k = 1, 2$ and $K_{A_k}(t_k, \omega_k) = \frac{1}{\sqrt{i2\pi b_k}} e^{i\left(\frac{a_k}{2b_k}t_k^2 - \frac{1}{b_k}t_k\omega_k + \frac{d_k}{2b_k}\omega_k^2\right)}$. The QLCT of $f \in L^1(\mathbb{R}^2; \mathbb{H})$ is given by [20,22]

$$L_{q,A}\{f\}(\omega_1, \omega_2) = \int_{\mathbb{R}^2} K_{i,A_1}(t_1, \omega_1) f(t_1, t_2) K_{j,A_2}(t_2, \omega_2) dt, \quad (18)$$

where $\mathbf{I} = (\mathbf{i}, \mathbf{j}), \mathbf{A} = (A_1, A_2)$ and

$$K_{\mathbf{I}, A_k}(t_k, \omega_k) = \frac{1}{\sqrt{\mathbf{I}_k 2\pi b_k}} e^{\mathbf{I}_k \left(\frac{a_k}{2b_k} t_k^2 - \frac{1}{b_k} t_k \omega_k + \frac{d_k}{2b_k} \omega_k^2 \right)}. \quad (19)$$

3. Generalized 2-D complex analytic signals

3.1. Derivation of parametric Hilbert transform

In this section, the 1-D parametric Hilbert transform (PHT) is derived by means of 2-D harmonic fields. A harmonic function $p(\mathbf{t})$ is a solution of Laplace equation

$$\nabla^2 p(\mathbf{t}) = \frac{\partial^2}{\partial t_1^2} p(\mathbf{t}) + \frac{\partial^2}{\partial t_2^2} p(\mathbf{t}) = 0, \quad (20)$$

where $\mathbf{t} = (t_1, t_2)^\top, \nabla = (\frac{\partial}{\partial t_1}, \frac{\partial}{\partial t_2})^\top$ [26,27]. Then we consider the harmonic potential field of $p(\mathbf{t})$ that is modulated by a complex exponential function in the t_1 direction

$$\mathbf{v}(\mathbf{t}) = (v_1(\mathbf{t}), v_2(\mathbf{t}))^\top = e^{-i\frac{a}{2b}t_1^2} \nabla p(\mathbf{t}). \quad (21)$$

It has practical significance in the study of wave propagation in media with spatially varying refractive indices, where it represents the complex gradient of the wavefront. We restrict the Laplace equation to $t_2 < 0$, with the boundary condition $v_2(t_1, 0) = f(t_1)$.

The harmonic potential field of $p(\mathbf{t})$ is irrotational and solenoidal in the open domain $t_2 < 0$. Therefore, the following equations hold

$$\text{rot}(\nabla p(\mathbf{t})) = \nabla \times p(\mathbf{t}) = \frac{\partial}{\partial t_1} e^{i\frac{a}{2b}t_1^2} v_1(\mathbf{t}) - \frac{\partial}{\partial t_2} e^{i\frac{a}{2b}t_1^2} v_2(\mathbf{t}) = 0, \quad (22)$$

$$\text{div}(\nabla p(\mathbf{t})) = \nabla \cdot p(\mathbf{t}) = \frac{\partial}{\partial t_1} e^{i\frac{a}{2b}t_1^2} v_1(\mathbf{t}) + \frac{\partial}{\partial t_2} e^{i\frac{a}{2b}t_1^2} v_2(\mathbf{t}) = 0. \quad (23)$$

Performing calculations in the linear canonical domain of time t_1 is easier, achieved by applying the 1D LCT. From (23) it follows that [see Appendix A]

$$-\frac{\omega_1^2}{b^2} L_A\{p\}(\omega_1, t_2) + \frac{\partial^2}{\partial t_2^2} L_A\{p\}(\omega_1, t_2) = 0. \quad (24)$$

Solving this equation for $t_3 < 0$ yields $L_A\{p\}(\omega_1, t_2) = C(\omega_1) e^{i\frac{\omega_1}{b}|t_2|^2}$, where $C(\omega_1)$ is independent of t_2 . Therefore, the $\mathbf{v}(\mathbf{t})$ in linear canonical domain has the form

$$\begin{aligned} e^{i\frac{a}{2b}t_1^2} L_A\{v_1\}(\omega_1, t_2) &= i\frac{\omega_1}{b} L_A\{p\}(\omega_1, t_2), \\ e^{i\frac{a}{2b}t_1^2} L_A\{v_2\}(\omega_1, t_2) &= \left| \frac{\omega_1}{b} \right| L_A\{p\}(\omega_1, t_2). \end{aligned} \quad (25)$$

Hence, $v_1(\mathbf{t})$ and $v_2(\mathbf{t})$ are a parametric Hilbert pair, which implies that the component $v_1(\mathbf{t})$ is the harmonic-conjugate of $v_2(\mathbf{t})$:

$$\begin{aligned} L_A\{v_1\}(\omega_1, t_2) &= \text{isgn}\left(\frac{\omega_1}{b}\right) L_A\{v_2\}(\omega_1, t_2) \\ &= H_p(\omega_1) L_A\{v_2\}(\omega_1, t_2). \end{aligned} \quad (26)$$

The analytic signal is defined as a complex function of real variable t_1 . For $t_2 \rightarrow 0^-$, we have

$$L_A\{v_1\}(\omega_1, 0) = H_p(\omega_1) L_A\{v_2\}(\omega_1, 0) = H_p(\omega_1) L_A\{f\}(\omega_1). \quad (27)$$

Applying the inverse LCT yields the analytic signal

$$\varphi(t_1) = v_2(t_1, 0) - \mathbf{i}v_1(t_1, 0) = f(t_1) - \mathbf{i}f_{H_p}(t_1), \quad (28)$$

where

$$f_{H_p}(t_1) = L_A^{-1}\{\text{isgn}\left(\frac{\omega_1}{b}\right) L_A\{f\}\} = -\frac{1}{\pi} \text{p.v.} \int \frac{f(\xi) e^{i\frac{a}{2b}(\xi^2 - t_1^2)}}{t_1 - \xi} d\xi. \quad (29)$$

3.2. Two-dimensional parametric Hilbert transform

In the previous section, it was demonstrated that the 1-D PHT can be derived by means of 2-D harmonic fields. For a 2-D signal $f(t_1, t_2)$, its partial parametric Hilbert transforms (PPHTs) can be defined directly as

$$f_{H_{p,P_1}}(t_1, t_2) = -\frac{1}{\pi} \text{p.v.} \int \frac{f(\xi_1, t_2)}{t_1 - \xi_1} e^{i\frac{a}{2b_1}(\xi_1^2 - t_1^2)} d\xi_1, \quad (30)$$

$$f_{H_{p,P_2}}(t_1, t_2) = -\frac{1}{\pi} \text{p.v.} \int \frac{f(t_1, \xi_2)}{t_2 - \xi_2} e^{i\frac{a_2}{2b_2}(\xi_2^2 - t_2^2)} d\xi_2. \quad (31)$$

And the total parametric Hilbert transform (TPHT) of f is defined by [22]

$$\begin{aligned} f_{H_{p,T}}(t_1, t_2) &= L_A^{-1}\{-\text{sgn}\left(\frac{\omega_1}{b_1}\right) \text{sgn}\left(\frac{\omega_2}{b_2}\right) L_A\{f(t_1, t_2)\}\} \\ &= \frac{1}{\pi^2} \text{p.v.} \iint \frac{f(\xi_1, \xi_2)}{(t_1 - \xi_1)(t_2 - \xi_2)} e^{i\frac{a_1}{2b_1}(\xi_1^2 - t_1^2)} e^{i\frac{a_2}{2b_2}(\xi_2^2 - t_2^2)} d\xi_1 d\xi_2, \end{aligned} \quad (32)$$

where $\mathbf{A} = (A_1, A_2), A_k = (a_k, b_k; c_k, d_k)$ with $a_k d_k - b_k c_k = 1, b_k \neq 0, k = 1, 2$.

3.3. Generalized 2-D complex analytic signals

The 2-D unit step linear canonical domain operators in the first and third quadrants of $(\omega_1/b_1, \omega_2/b_2)$ are defined as

$$\begin{aligned} 4\mathbf{I}(\omega_1/b_1, \omega_2/b_2) &= [1 + \text{sgn}(\omega_1/b_1)] [1 + \text{sgn}(\omega_2/b_2)], \\ 4\mathbf{I}(-\omega_1/b_1, -\omega_2/b_2) &= [1 - \text{sgn}(\omega_1/b_1)] [1 - \text{sgn}(\omega_2/b_2)]. \end{aligned} \quad (33)$$

The inverse 2-D LCT yields the generalized 2-D complex analytic signals (GCSs) with single-quadrant spectrum of the form

$$\begin{aligned} \varphi_{G,1}(\mathbf{t}) &= f(\mathbf{t}) - f_{H_{p,T}}(\mathbf{t}) - \mathbf{i} [f_{H_{p,P_1}}(\mathbf{t}) + f_{H_{p,P_2}}(\mathbf{t})], \\ \varphi_{G,2}(\mathbf{t}) &= f(\mathbf{t}) - f_{H_{p,T}}(\mathbf{t}) + \mathbf{i} [f_{H_{p,P_1}}(\mathbf{t}) + f_{H_{p,P_2}}(\mathbf{t})], \end{aligned} \quad (34)$$

where $\mathbf{t} = (t_1, t_2)^\top$, and $f_{H_{p,P_1}}, f_{H_{p,P_2}}$ are the PPHTs of f which are defined in (30) and (31), and $f_{H_{p,T}}$ is the TPHT defined in (32). Here, a complex analytic signal with a spectrum confined to a single quadrant within a linear canonical domain is defined.



Fig. 1. (a) The 2-D Gaussian signal $f(t_1, t_2)$ in (35). (b) The spectrogram of $f(t_1, t_2)$.

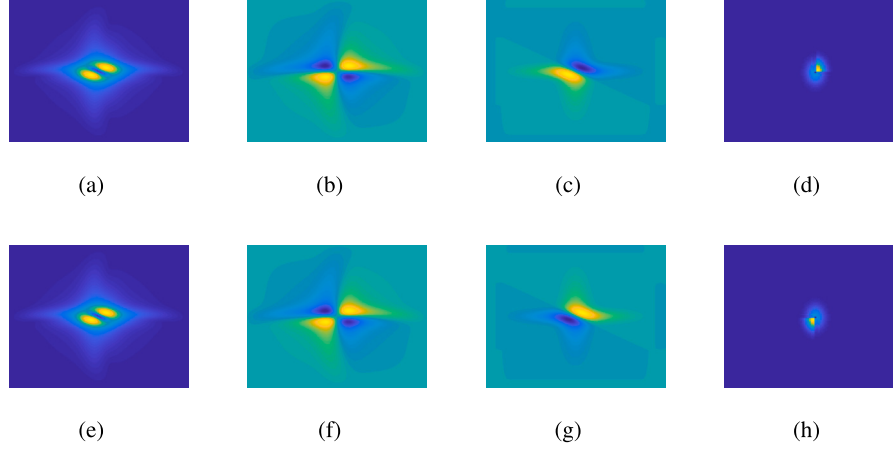


Fig. 2. The GCSs $\varphi_{G,1}(\mathbf{t}), \varphi_{G,2}(\mathbf{t})$ of the 2-D Gaussian signal $f(\mathbf{t})$ in (35). (a) The modulus of $\varphi_{G,1}$; (b) The real part of $\varphi_{G,1}$; (c) The imaginary part of $\varphi_{G,1}$; (d) The spectrogram of $\varphi_{G,1}$; (e) The modulus of $\varphi_{G,2}$; (f) The real part of $\varphi_{G,2}$; (g) The imaginary part of $\varphi_{G,2}$; (h) The spectrogram of $\varphi_{G,2}$.

Consider a 2-D Gaussian signal of the form

$$f(t_1, t_2) = e^{-\pi \left(\frac{t_1^2}{0.5^2} + \frac{t_2^2}{0.3^2} \right)}. \quad (35)$$

Fig. 1 displays the time-domain and frequency-domain diagrams of the signal (35), and the GCSs of it with $A_1 = (1/2, 1/2; -2, 0)$, $A_2 = (1/2, 1/2; -2, 0)$ are shown in Fig. 2. Figs. 2 (d) and (h) demonstrate that the linear canonical spectrum of the GCSs of the signal is predominantly concentrated in the first and third quadrants, respectively.

4. Generalized 2-D quaternionic analytic signals

In the previous section, a generalized analytic signal in the complex domain was defined using PPHTs and TPHT. However, with the advancements in quaternion analysis tools, it is of significant importance to define generalized analytic signals in the quaternion domain. Analytic signals in the quaternion domain offer greater degrees of freedom and expressive power, thereby enabling enhanced processing of multi-dimensional data.

In this section, we introduce the definition of a generalized 2-D analytical signal in the quaternion domain, referred to as the generalized 2-D quaternion analytic signal (GQS) [22]. As shown in (6), the quaternion linear canonical spectrum in the first quadrant is given by $4\mathbf{I}(\omega_1/b_1, \omega_2/b_2)L_{q,A}(\omega_1, \omega_2)$. Let $\mathbf{t} = (t_1, t_2)^\top$, the inverse QLCT of this spectrum yields the GQS with single-quadrant spectrum of the form

$$\varphi_{G,Q}(\mathbf{t}) = f(\mathbf{t}) - \mathbf{i}f_{H_p,P_1}(\mathbf{t}) - \mathbf{j}f_{H_p,P_2}(\mathbf{t}) + \mathbf{k}f_{H_p,T}(\mathbf{t}), \quad (36)$$

where $f_{H_p,P_1}, f_{H_p,P_2}, f_{H_p,T}$ are the same as used in (34). The conjugate of $\varphi_{G,Q}(\mathbf{t})$ is

$$\varphi_{G,Q}^*(\mathbf{t}) = f(\mathbf{t}) + \mathbf{i}f_{H_p,P_1}(\mathbf{t}) + \mathbf{j}f_{H_p,P_2}(\mathbf{t}) - \mathbf{k}f_{H_p,T}(\mathbf{t}). \quad (37)$$

Here is a simple example to illustrate this concept. The GQS of the signal (35) with $A_1 = (0, 98/99; -99/98, 0)$, $A_2 = (0, 98/99; -99/98, 0)$ and its spectrum are shown in Fig. 3. Figs. 3 (a-d) depict the GQS generated by the signal (35), revealing distinct local directional properties with varying imaginary units. Figs. 3 (e-h) display the corresponding spectrum of these GQS, exhibiting concentrated energy within a single quadrant.

5. Generalized 2-D monogenic signals

5.1. Parametric Riesz transform

Although the 2-D PHTs (PPHTs, TPHT) are defined in section 3.2, they are unable to perform a phase shift of $\pi/2$, resulting in the analytic signals defined by these equations being anisotropic functions. Therefore, the use of “isotropic 2-D parametric Hilbert transforms,” known as parametric Riesz transforms (PRTs), is required. Similar to the PHT, which can be derived from a 2-D harmonic field, the PRTs can be derived from a 3-D harmonic fields.

Let $p(\mathbf{t})$ be the solution of Laplace equation $\nabla^2 p(\mathbf{t}) = \frac{\partial^2}{\partial t_1^2} p(\mathbf{t}) + \frac{\partial^2}{\partial t_2^2} p(\mathbf{t}) + \frac{\partial^2}{\partial t_3^2} p(\mathbf{t}) = 0$, where $\mathbf{t} = (t_1, t_2, t_3)^\top$. Denoting the harmonic potential field of $p(\mathbf{t})$ modulated by a complex exponential function in the t_1 and t_2 directions

$$\mathbf{v}(\mathbf{t}) = (v_1(\mathbf{t}), v_2(\mathbf{t}), v_3(\mathbf{t}))^\top = e^{-\mathbf{i}\frac{a_1}{2b_1}t_1^2} e^{-\mathbf{i}\frac{a_2}{2b_2}t_2^2} \nabla p(\mathbf{t}), \quad (38)$$

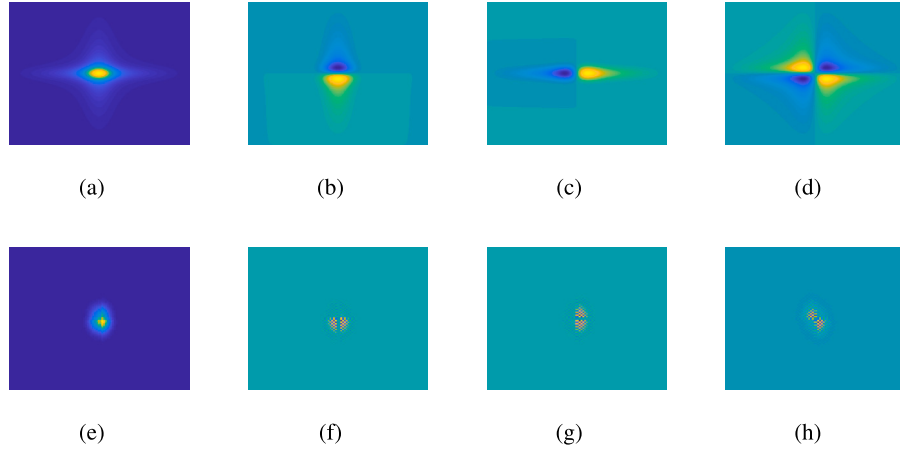


Fig. 3. The GQS $\varphi_{G,Q}(\mathbf{t})$ of the 2-D Gaussian signal $f(\mathbf{t})$ in (35) and its spectrum. (a) The modulus of $\varphi_{G,Q}$; (b) i-part of $\varphi_{G,Q}$; (c) j-part of $\varphi_{G,Q}$; (d) k-part of $\varphi_{G,Q}$; (e) The modulus of the spectrum; (f) i-part of the spectrum; (g) j-part of the spectrum; (h) k-part of the spectrum.

where $\nabla = (\frac{\partial}{\partial t_1}, \frac{\partial}{\partial t_2}, \frac{\partial}{\partial t_3})^\top$, and the boundary condition is $v_3(t_1, t_2, 0) = f(t_1, t_2)$. The harmonic potential field of $p(\mathbf{t})$ is irrotational and solenoidal in the open domain $t_3 < 0$, that is

$$\text{rot}(\nabla p(\mathbf{t})) = \nabla \times p(\mathbf{t}) = 0, \quad (39)$$

$$\begin{aligned} \text{div}(\nabla p(\mathbf{t})) = \nabla \cdot p(\mathbf{t}) &= \frac{\partial}{\partial t_1} e^{\frac{a_1}{2b_1} t_1^2} e^{\frac{a_2}{2b_2} t_2^2} v_1 \\ &+ \frac{\partial}{\partial t_2} e^{\frac{a_1}{2b_1} t_1^2} e^{\frac{a_2}{2b_2} t_2^2} v_2 + \frac{\partial}{\partial t_3} e^{\frac{a_1}{2b_1} t_1^2} e^{\frac{a_2}{2b_2} t_2^2} v_3 = 0. \end{aligned} \quad (40)$$

Applying the 2D LCT on both sides of (40), it follows that

$$-\left(\sqrt{\frac{\omega_1^2}{b_1^2} + \frac{\omega_2^2}{b_2^2}}\right)^2 L_A\{p\}(\omega_1, \omega_2, t_3) + \frac{\partial^2}{\partial t_3^2} L_A\{p\}(\omega_1, \omega_2, t_3) = 0. \quad (41)$$

Then $L_A\{p\}(\omega_1, \omega_2, t_3) = C(\omega_1, \omega_2) e^{\sqrt{\frac{\omega_1^2}{b_1^2} + \frac{\omega_2^2}{b_2^2}} t_3}$ ($t_3 < 0$), where $C(\omega_1, \omega_2)$ is independent of t_3 . The relations between $L_A\{v_1\}(\omega_1, \omega_2, t_3)$, $L_A\{v_2\}(\omega_1, \omega_2, t_3)$, and $L_A\{v_3\}(\omega_1, \omega_2, t_3)$ is as follows

$$L_A\{v_1\}(\omega_1, \omega_2, t_3) = \mathbf{i} \frac{\omega_1/b_1}{\sqrt{(\omega_1/b_1)^2 + (\omega_2/b_2)^2}} L_A\{v_3\}(\omega_1, \omega_2, t_3), \quad (42)$$

$$L_A\{v_2\}(\omega_1, \omega_2, t_3) = \mathbf{j} \frac{\omega_2/b_2}{\sqrt{(\omega_1/b_1)^2 + (\omega_2/b_2)^2}} L_A\{v_3\}(\omega_1, \omega_2, t_3). \quad (43)$$

Denoting $\boldsymbol{\omega} = (\omega_1, \omega_2)^\top$, $\mathbf{b} = (b_1, b_2)^\top$, $\boldsymbol{\omega}/\mathbf{b} = (\omega_1/b_1, \omega_2/b_2)^\top$ and letting $R_p(\omega_1, \omega_2) = (R_{p,1}(\omega_1, \omega_2), R_{p,2}(\omega_1, \omega_2))^\top = (L_A\{v_1\}(\omega_1, \omega_2, 0), L_A\{v_2\}(\omega_1, \omega_2, 0))^\top$, it follows that the relationship also holds for the continuous extension of $\mathbf{v}$ for $t_3 \rightarrow 0^-$. Therefore, the Riesz transformed signal in the linear canonical domain is defined as

$$R_p(\boldsymbol{\omega}) = (R_{p,1}(\boldsymbol{\omega}), R_{p,2}(\boldsymbol{\omega}))^\top = \mathbf{i} \frac{\boldsymbol{\omega}/\mathbf{b}}{|\boldsymbol{\omega}/\mathbf{b}|} L_A\{f\}(\boldsymbol{\omega}). \quad (44)$$

In the following we use the notation $\mathbf{t} = (t_1, t_2)^\top$ since $t_3 = 0$. The multiplication in the linear canonical domain in (44) corresponds to the following convolution in the spatial domain

$$\begin{aligned} f_{R_p}(\mathbf{t}) &= (f_{R_{p,1}}(\mathbf{t}), f_{R_{p,2}}(\mathbf{t}))^\top = L_A^{-1}\{R_p(\boldsymbol{\omega})\} \\ &= e^{-\frac{a_1}{2b_1} t_1^2} e^{-\frac{a_2}{2b_2} t_2^2} \left[f(\mathbf{t}) e^{\frac{a_1}{2b_1} t_1^2} e^{\frac{a_2}{2b_2} t_2^2} \ast \frac{\mathbf{t}}{2\pi|\mathbf{t}|^3} \right] \\ &= \frac{1}{2\pi} \text{p.v.} \iint \frac{f(\boldsymbol{\eta}) e^{\frac{a_1}{2b_1} (t_1^2 - \eta_1^2)} e^{\frac{a_2}{2b_2} (t_2^2 - \eta_2^2)}}{|\mathbf{t} - \boldsymbol{\eta}|^3} d\eta_1 d\eta_2, \end{aligned} \quad (45)$$

where $\boldsymbol{\eta} = (\eta_1, \eta_2)^\top$. This is the temporal expression of PRT.

5.2. Generalized 2-D monogenic signals

The algebra of complex numbers is insufficient to embed a GAS based on PRT, which consists of the signal $f(\mathbf{t}), \mathbf{t} = (t_1, t_2)^\top$, and its PRT vector $f_{R_p}(\mathbf{t})$, with a total of three components. Actually, function theory corresponding to 3-D vector fields is given by means of Clifford analysis. Therefore, the generalized 2-D monogenic signal (GMS) in the quaternionic domain using the PRT is defined by

$$\begin{aligned} M_p(\boldsymbol{\omega}) &= L_A\{v_3\}(\omega_1, \omega_2, 0) - \mathbf{i} L_A\{v_1\}(\omega_1, \omega_2, 0) - \mathbf{j} L_A\{v_2\}(\omega_1, \omega_2, 0) \\ &= L_A\{f\}(\boldsymbol{\omega}) - \mathbf{i} R_{p,i,1}(\boldsymbol{\omega}) - \mathbf{j} R_{p,j,2}(\boldsymbol{\omega}), \end{aligned} \quad (46)$$

which is equivalent to

$$\varphi_{G,M}(\mathbf{t}) = f(\mathbf{t}) - \mathbf{i} f_{R_{p,i,1}}(\mathbf{t}) - \mathbf{j} f_{R_{p,j,2}}(\mathbf{t}). \quad (47)$$

The conjugate of $\varphi_{G,M}(\mathbf{t})$ is $\varphi_{G,M}^*(\mathbf{t}) = f(\mathbf{t}) + \mathbf{i} f_{R_{p,i,1}}(\mathbf{t}) + \mathbf{j} f_{R_{p,j,2}}(\mathbf{t})$. The GMS of the signal (35) with $A_1 = (0, 1; -1, 0)$, $A_2 = (0, 1; -1, 0)$ and its spectrum are shown in Fig. 4. Figs. 4 (d-f) display the directional spectrum corresponding to the GMS.

6. Four-dimensional WDLs and AFLs of generalized 2-D analytic signals

In the previous sections, three definitions of GAS were introduced, namely GCS, GQS, and GMS. In this section, we delve into the applications of these three types of GASs in time-frequency analysis, considering their 4-D WDLs.

6.1. Four-dimensional WDL and AFL of GCSs

Two different generalized complex analytic signals $\varphi_{G,1}(\mathbf{t})$ and $\varphi_{G,2}(\mathbf{t})$ are defined by (34). The 4-D WDLs and AFLs of these signals are defined by the 2-D LCT of correlation products

$$r_{G,k}(t_1, t_2, \tau_1, \tau_2) = \varphi_{G,k}(t_1 + \frac{\tau_1}{2}, t_2 + \frac{\tau_2}{2}) \varphi_{G,k}^*(t_1 - \frac{\tau_1}{2}, t_2 - \frac{\tau_2}{2}), \quad (48)$$

where $k = 1, 2$.

Definition 1. The 4-D WDLs and AFLs of the GCSs $\varphi_{G,1}(\mathbf{t})$ and $\varphi_{G,2}(\mathbf{t})$ are defined by

$$\begin{aligned} W_{\hat{A},l-2}(t_1, t_2, \omega_1, \omega_2) \\ &= \iint r_{G,l-2}(t_1, t_2, \tau_1, \tau_2) K_{i,A_3}(\tau_1, \omega_1) K_{i,A_4}(\tau_2, \omega_2) d\tau_1 d\tau_2, \\ AF_{\hat{A},l-2}(\tau_1, \tau_2, \omega_1, \omega_2) \end{aligned} \quad (49)$$

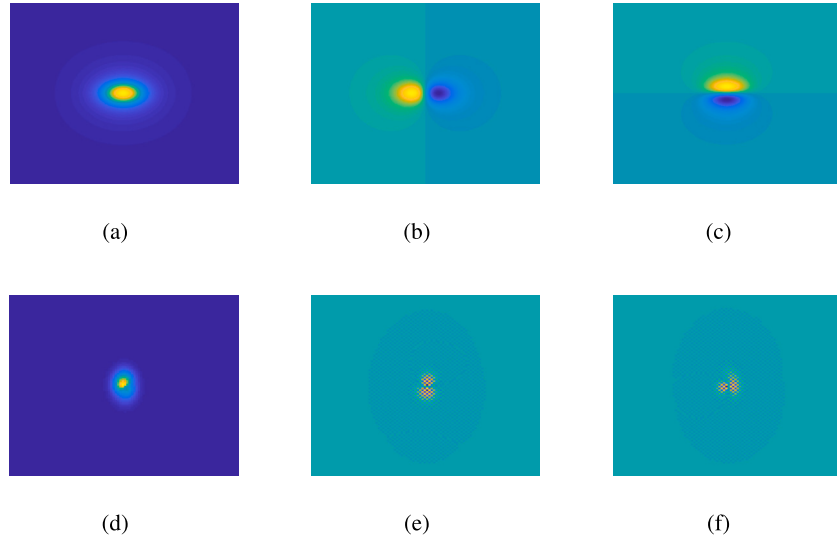


Fig. 4. The GMS $\varphi_{G,M}(t)$ of the 2-D Gaussian signal $f(t)$ in (35) and its spectrum. (a) The modulus of $\varphi_{G,M}$; (b) i-part of $\varphi_{G,M}$; (c) j-part of $\varphi_{G,M}$; (d) The modulus of the spectrum; (e) i-part of the spectrum; (f) j-part of the spectrum.

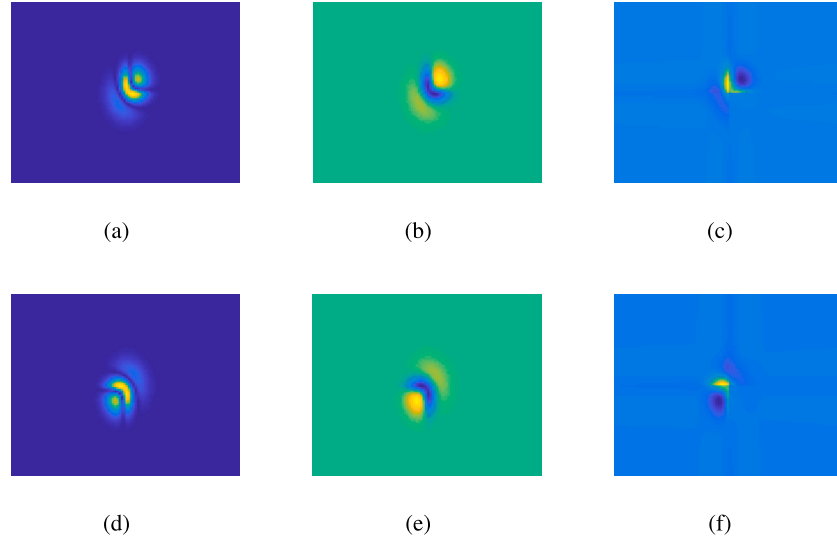


Fig. 5. The 4-D WDLs $W_{\hat{\Lambda},1}(t_1, t_2, \omega_1, \omega_2)$, $W_{\hat{\Lambda},2}(t_1, t_2, \omega_1, \omega_2)$ for the GCSs of signal (35). (a) Modulus of $W_{\hat{\Lambda},1}$; (b) Real-part of $W_{\hat{\Lambda},1}$; (c) Imaginary-part of $W_{\hat{\Lambda},1}$; (d) Modulus of $W_{\hat{\Lambda},2}$; (e) Real-part of $W_{\hat{\Lambda},2}$; (f) Imaginary-part of $W_{\hat{\Lambda},2}$.

$$= \iint r_{G,l-2}(t_1, t_2, \tau_1, \tau_2) K_{i,A_3}(t_1, \omega_1) K_{i,A_4}(t_2, \omega_2) dt_1 dt_2, \quad (50)$$

where $\hat{\mathbf{A}} = (A_3, A_4)$, $A_l = (a_l, b_l; c_l, d_l)$, $l = 3, 4$ with $a_l d_l - b_l c_l = 1$, $b_l \neq 0$, and $K_{i,A_l}(t_{l-2}, \omega_{l-2}) = \frac{1}{\sqrt{12\pi b_l}} e^{i\left(\frac{a_l}{2b_l} t_{l-2}^2 - \frac{1}{b_l} t_{l-2} \omega_{l-2} + \frac{d_l}{2b_l} \omega_{l-2}^2\right)}$.

The 4-D WDLs $W_{\hat{\Lambda},1}(t_1, t_2, \omega_1, \omega_2)$, $W_{\hat{\Lambda},2}(t_1, t_2, \omega_1, \omega_2)$ for the GCSs of (35) with $A_1 = A_2 = (0.4, 1/2; -2, 0)$, $A_3 = A_4 = (0, 0.9999; -1.0001, 0)$ are shown in Fig. 5.

6.2. Four-dimensional WDL and AFL of GQS

The 4-D WDLs and AFLs of the GQS $\varphi_{G,Q}(t)$ are defined by QLCT of quaternionic correlation product

$$r_{G,Q}(t_1, t_2, \tau_1, \tau_2) = \varphi_{G,Q}(t_1 + \frac{\tau_1}{2}, t_2 + \frac{\tau_2}{2}) \varphi_{G,Q}^*(t_1 - \frac{\tau_1}{2}, t_2 - \frac{\tau_2}{2}). \quad (51)$$

The insertion of the quaternionic signals (36) and (37) in (51) yields the following form of the quaternionic correlation product

$$r_{G,Q}(t_1, t_2, \tau_1, \tau_2) = A_Q + \mathbf{i}B_Q + \mathbf{j}C_Q + \mathbf{k}D_Q, \quad (52)$$

with

$$\begin{aligned} A_Q &= f^+ f^- + f_{H_p, P_1}^+ f_{H_p, P_1}^- + f_{H_p, P_2}^+ f_{H_p, P_2}^- + f_{H_p, T}^+ f_{H_p, T}^-, \\ B_Q &= f^+ f_{H_p, P_1}^- - f_{H_p, P_1}^+ f^- + f_{H_p, P_2}^+ f_{H_p, T}^- - f_{H_p, T}^+ f_{H_p, P_2}^-, \\ C_Q &= f^+ f_{H_p, P_2}^- - f_{H_p, P_1}^+ f_{H_p, T}^- - f_{H_p, P_2}^+ f^- + f_{H_p, T}^+ f_{H_p, P_1}^-, \\ D_Q &= -f^+ f_{H_p, T}^- - f_{H_p, P_1}^+ f_{H_p, P_2}^- + f_{H_p, P_2}^+ f_{H_p, P_1}^- + f_{H_p, T}^+ f^-, \end{aligned} \quad (53)$$

where $f^+ = f(t_1 + \frac{\tau_1}{2}, t_2 + \frac{\tau_2}{2})$, $f^- = f(t_1 - \frac{\tau_1}{2}, t_2 - \frac{\tau_2}{2})$ and the same for the other functions f_{H_p, P_1}^+ , f_{H_p, P_2}^+ , $f_{H_p, T}^+$.



Fig. 6. The 4-D WDL $W_{\hat{A},Q}(0,0,\omega_1,\omega_2)$ of generalized 2-D quaternionic signal of (35). (a) Real-part of $W_{\hat{A},Q}$; (b) i-part of $W_{\hat{A},Q}$; (c) j-part of $W_{\hat{A},Q}$; (d) The k-part of $W_{\hat{A},Q}$.

Definition 2. The 4-D WDLs and AFLs of the GQS $\varphi_{G,Q}(\mathbf{t})$ are given by

$$W_{\hat{A},Q}(t_1, t_2, \omega_1, \omega_2) = \iint K_{i,A_3}(\tau_1, \omega_1) r_{G,Q}(t_1, t_2, \tau_1, \tau_2) K_{j,A_4}(\tau_2, \omega_2) d\tau_1 d\tau_2, \quad (54)$$

$$AF_{\hat{A},Q}(\tau_1, \tau_2, \omega_1, \omega_2) = \iint K_{i,A_3}(t_1, \omega_1) r_{G,Q}(t_1, t_2, \tau_1, \tau_2) K_{j,A_4}(t_2, \omega_2) dt_1 dt_2, \quad (55)$$

where $\hat{A} = (A_3, A_4)$, $A_l = (a_l, b_l; c_l, d_l)$, $l = 3, 4$ with $a_l d_l - b_l c_l = 1$, $b_l \neq 0$, and $K_{I_{l-2}, A_l}(t_{l-2}, \omega_{l-2})$, $\mathbf{I} = (\mathbf{i}, \mathbf{j})$ are equivalent to (19).

The insertion of (52) in (54) and (55) yield the following form of $W_{\hat{A},Q}$ and $AF_{\hat{A},Q}$

$$W_{\hat{A},Q}(t_1, t_2, \omega_1, \omega_2) = W_{\hat{A},Qr} + \mathbf{i}W_{\hat{A},Qi} + \mathbf{j}W_{\hat{A},Qj} + \mathbf{k}W_{\hat{A},Qk},$$

$$AF_{\hat{A},Q}(\tau_1, \tau_2, \omega_1, \omega_2) = AF_{\hat{A},Qr} + \mathbf{i}AF_{\hat{A},Qi} + \mathbf{j}AF_{\hat{A},Qj} + \mathbf{k}AF_{\hat{A},Qk}, \quad (56)$$

where the related terms are given in Appendix B.

The 4-D WDL $W_{\hat{A},Q}(t_1, t_2, \omega_1, \omega_2)$ of GQS of (35) with $A_1 = A_2 = (0, 98/99; -99/98, 0)$, $A_3 = A_4 = (0, 0.9999; -1.0001, 0)$ is shown in Fig. 6.

6.3. Four-dimensional WDL and AFL of generalized 2-D monogenic signals

The WDL and AFL of GMS (47) are defined by QLCT of monogenic correlation product

$$r_{G,M}(t_1, t_2, \tau_1, \tau_2) = \varphi_{G,M}(t_1 + \frac{\tau_1}{2}, t_2 + \frac{\tau_2}{2}) \varphi_{G,M}^*(t_1 - \frac{\tau_1}{2}, t_2 - \frac{\tau_2}{2}). \quad (57)$$

The insertion of the GMS (47) in (57) yields

$$r_{G,M}(t_1, t_2, \tau_1, \tau_2) = A_M + \mathbf{i}B_M + \mathbf{j}C_M + \mathbf{k}D_M, \quad (58)$$

where

$$\begin{aligned} A_M &= f^+ f^- + f_{R_{p,i,1}}^+ f_{R_{p,i,1}}^- + f_{R_{p,j,2}}^+ f_{R_{p,j,2}}^-, \\ B_M &= f^+ f_{R_{p,i,1}}^- - f_{R_{p,i,1}}^+ f^-, \\ C_M &= f^+ f_{R_{p,j,2}}^- - f_{R_{p,j,2}}^+ f^-, \\ D_M &= -f_{R_{p,i,1}}^+ f_{R_{p,j,2}}^- + f_{R_{p,j,2}}^+ f_{R_{p,i,1}}^-, \end{aligned} \quad (59)$$

with $f^+ = f(t_1 + \frac{\tau_1}{2}, t_2 + \frac{\tau_2}{2})$, $f^- = f(t_1 - \frac{\tau_1}{2}, t_2 - \frac{\tau_2}{2})$ and the same for the other functions $f_{R_{p,i,1}}$, $f_{R_{p,j,2}}$.

Definition 3. The 4-D WDLs and AFLs of the GMS $\varphi_{G,M}(\mathbf{t})$ are given by

$$W_{\hat{A},M}(t_1, t_2, \omega_1, \omega_2) = \iint K_{i,A_1}(\tau_1, \omega_1) r_{G,M}(t_1, t_2, \tau_1, \tau_2) K_{j,A_2}(\tau_2, \omega_2) d\tau_1 d\tau_2, \quad (60)$$

$$AF_{\hat{A},M}(\tau_1, \tau_2, \omega_1, \omega_2)$$

$$= \iint K_{i,A_1}(t_1, \omega_1) r_{G,M}(t_1, t_2, \tau_1, \tau_2) K_{j,A_2}(t_2, \omega_2) dt_1 dt_2, \quad (61)$$

where $\hat{A} = (A_3, A_4)$, $A_l = (a_l, b_l; c_l, d_l)$, $l = 3, 4$ with $a_l d_l - b_l c_l = 1$, $b_l \neq 0$, and $K_{I_{l-2}, A_l}(t_{l-2}, \omega_{l-2})$, $\mathbf{I} = (\mathbf{i}, \mathbf{j})$ are equals to (19).

Furthermore, the $W_{\hat{A},M}$ and $AF_{\hat{A},M}$ are also quaternion-valued function of the form (56) based on (58)

$$W_{\hat{A},M}(t_1, t_2, \omega_1, \omega_2) = W_{\hat{A},Mr} + \mathbf{i}W_{\hat{A},Mi} + \mathbf{j}W_{\hat{A},Mj} + \mathbf{k}W_{\hat{A},Mk},$$

$$AF_{\hat{A},M}(\tau_1, \tau_2, \omega_1, \omega_2) = AF_{\hat{A},Mr} + \mathbf{i}AF_{\hat{A},Mi} + \mathbf{j}AF_{\hat{A},Mj} + \mathbf{k}AF_{\hat{A},Mk}. \quad (62)$$

The 4-D WDL $W_{\hat{A},M}(t_1, t_2, \omega_1, \omega_2)$ of GMS of (35) with $A_1 = A_2 = (0, 1; -1, 0)$, $A_3 = A_4 = (0, 0.9999; -1.0001, 0)$ is shown in Fig. 7.

7. Potential applications

7.1. Envelope detection using generalized 2-D analytic signals

In this section, the applications of the three previously discussed GASSs will be explored. These three types of GASSs effectively suppress the negative frequencies in the LCT of the original signal. Additionally, due to the affine properties introduced by the additional parameters of the LCT itself, these GASSs exhibit a wider range of application scenarios and demonstrate more diverse effects in practical settings. Similar to the polar decomposition of analytical signals, the local amplitude of GCSs, GQS and GMS can be obtained by

$$\begin{aligned} \|\varphi_{G,1}(\mathbf{t})\| &= \|\varphi_{G,2}(\mathbf{t})\| = \sqrt{(f - f_{H_p,T})^2 + (f_{H_p,P_1} + f_{H_p,P_2})^2}, \\ \|\varphi_{G,Q}(\mathbf{t})\| &= \sqrt{f^2(\mathbf{t}) + \|f_{H_p}(\mathbf{t})\|^2}, \\ \|\varphi_{G,M}(\mathbf{t})\| &= \sqrt{f^2(\mathbf{t}) + \|f_{R_A}(\mathbf{t})\|^2}, \end{aligned} \quad (63)$$

where $\|f_{H_p}\| = \sqrt{f_{H_p,P_1}^2(\mathbf{t}) + f_{H_p,P_2}^2(\mathbf{t}) + f_{H_p,T}^2(\mathbf{t})}$, $\|f_{R_A}(\mathbf{t})\| = \sqrt{f_{R_{A,i,1}}^2(\mathbf{t}) + f_{R_{A,j,2}}^2(\mathbf{t})}$. It can be observed that the local amplitude of these generalized 2-D analytic signals represents the spatial envelope of the original image. By squaring the local amplitude, the local energy of the image can be obtained. Therefore, these generalized 2-D analytical signals can be used for image edge detection.

Below, the practical application effects of the three types of GASSs will be demonstrated using the Cameraman and Peppers images as examples. The parameter $(0, 3; -3, 0)$ is employed in all of the subsequent GASSs. Figs. 8 (a) and (d) show the results of GCSs on Cameraman images. The real and imaginary parts of GCSs exhibit different local directional properties, respectively. However, the final edge detection performance of GCS was poor, and the image appeared relatively blurry. Fig. 9 and 10 (a) show the edge detection results of GQS and GMS, respectively, and they can be seen that both methods exhibit good image features.

Furthermore, building upon the excellent performance of GMS, we propose an envelope detection method that leverages GMS to opti-



Fig. 7. The 4-D WDL $W_{\hat{A},M}(0,0,\omega_1,\omega_2)$ of generalized 2-D monogenic signal of (35). (a) Real-part of $W_{\hat{A},M}$; (b) i-part of $W_{\hat{A},M}$; (c) j-part of $W_{\hat{A},M}$; (d) The k-part of $W_{\hat{A},M}$.

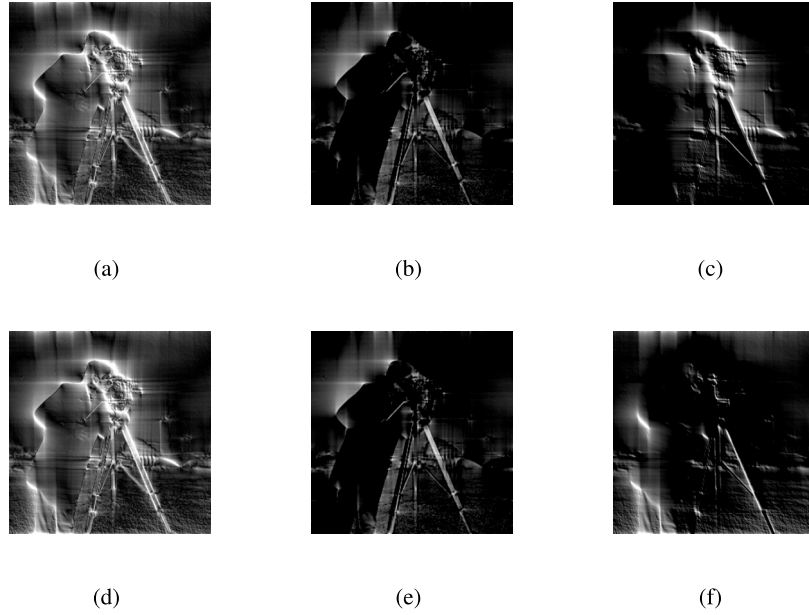


Fig. 8. The GCSs $\varphi_{G,1}(\mathbf{t})$, $\varphi_{G,2}(\mathbf{t})$ of Cameraman image. (a) The modulus of $\varphi_{G,1}$; (b) The real-part of $\varphi_{G,1}$; (c) The imaginary-part of $\varphi_{G,1}$; (d) The modulus of $\varphi_{G,2}$; (e) The real-part of $\varphi_{G,2}$; (f) The imaginary-part of $\varphi_{G,2}$.



Fig. 9. The GQS $\varphi_{G,Q}(\mathbf{t})$ of Cameraman image. (a) The modulus of $\varphi_{G,Q}$; (b) i-part of $\varphi_{G,Q}$; (c) j-part of $\varphi_{G,Q}$; (d) The k-part of $\varphi_{G,Q}$.

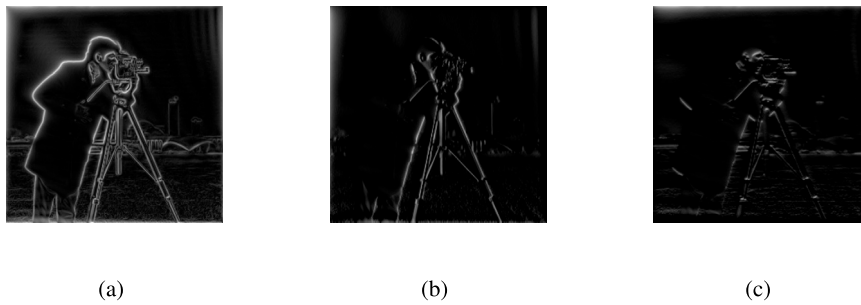


Fig. 10. The GMS $\varphi_{G,M}(\mathbf{t})$ of Cameraman image. (a) The modulus of $\varphi_{G,M}$; (b) i-part of $\varphi_{G,M}$; (c) j-part of $\varphi_{G,M}$.

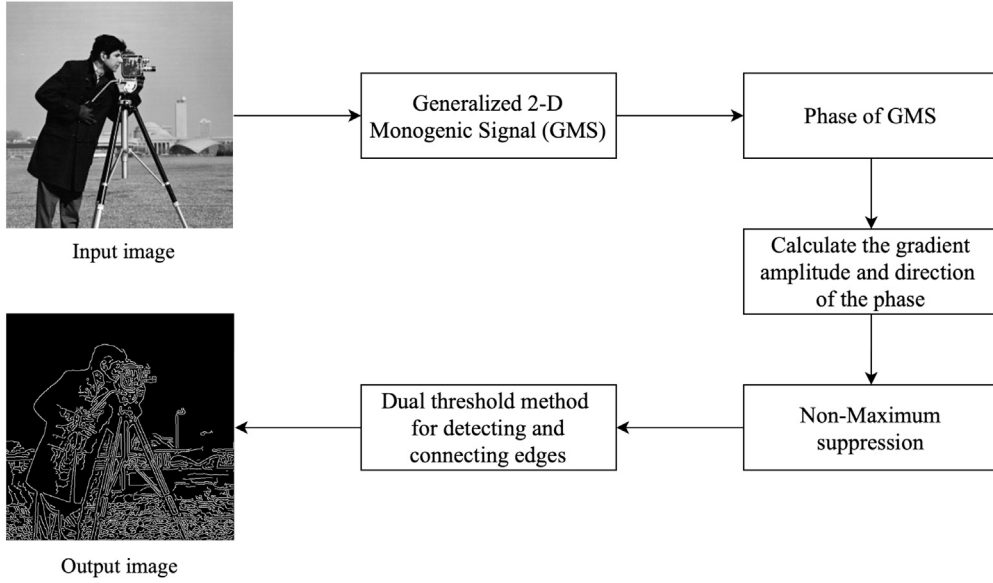


Fig. 11. Algorithm flowchart.

minimize its performance. The algorithm flowchart is depicted in Fig. 11. For an input image, firstly compute its GMS based on the content described in Section 5. Following that, calculate the phase by $\Phi(\mathbf{t}) = \arctan\left(\frac{\sqrt{f_{R_{p,1}}^2(\mathbf{t}) + f_{R_{p,2}}^2(\mathbf{t})}}{f(\mathbf{t})}\right)$, $\mathbf{t} = (t_1, t_2)$. Next, employ a finite difference method with 2×2 neighborhood first-order partial derivatives, calculate the gradient amplitude $M(\mathbf{t})$ and direction $D(\mathbf{t})$ of $\Phi(\mathbf{t})$, i.e. $M(\mathbf{t}) = \sqrt{k_{t_1}^2(\mathbf{t}) + k_{t_2}^2(\mathbf{t})}$, $D(\mathbf{t}) = \arctan\left(\frac{k_{t_1}(\mathbf{t})}{k_{t_2}(\mathbf{t})}\right)$, where k_{t_1} and k_{t_2} represent the results of filtering $\Phi(\mathbf{t})$ with filters $\rho_{t_1} = (-0.5, 0.5; -0.5, 0.5)$ and $\rho_{t_2} = (0.5, 0.5; -0.5, -0.5)$ along the rows and columns, respectively. Subsequently, conduct non-maximum suppression on the gradient amplitude. Finally, the dual threshold method is used to detect and connect edges to obtain the final detection result [28]. We have selected three commonly used operators, namely Sobel, Roberts, and Canny, for comparison, as illustrated in Fig. 12. Let $A_1 = A_2 = (0, 3; -3, 0)$, it is evident that our method exhibits excellent performance in regions with indistinct edges, particularly in dark areas.

7.2. Feature extraction using 4-D WDLs

In this section, the potential application of 4-D WDLs in GASs is discussed. Consider Newton's rings, which are circular interference patterns generated by optical interference [29,30]. The model of a Newton's ring with a physical size of T_N is defined as follows

$$h_1(x, y) = \text{rect}(x/T_N) \text{rect}(y/T_N) I_0, \quad (64)$$

$$h_2(x, y) = -\frac{I_1}{2} \text{rect}(x/T_N) \exp\left[j\pi \frac{2}{\lambda R} x^2\right] \text{rect}(y/T_N) \exp\left[j\pi \frac{2}{\lambda R} y^2\right], \quad (65)$$

$$h_3(x, y) = -\frac{I_1}{2} \text{rect}(x/T_N) \exp\left[-j\pi \frac{2}{\lambda R} x^2\right] \text{rect}(y/T_N) \exp\left[-j\pi \frac{2}{\lambda R} y^2\right], \quad (66)$$

where I_0 represents the mean intensity, $I_1(x, y) = h_1(x, y) + h_2(x, y) + h_3(x, y)$ is the amplitude of the fringes, λ represents the wavelength of the light, and R represents the curvature radius of the optical lens. Let $T_N = 256$, $\lambda = 589.3$, $R = 5.1$, then the Newton's ring as shown in Fig. 13.

We calculate the GCS, GQS, and GMS of the Newton's ring depicted in Fig. 13. Subsequently, we compute the 4-D WDLs of these three GASs, as illustrated in Fig. 14. Figs. 14 (a), (d), and (h) show the magnitudes of the 4-D WDLs for the GCS, GQS, and GMS of Newton's rings, respectively. This method effectively captures the characteristics of Newton's rings, resulting in significant spectral impulse.

Then we apply the above method to Lena images and select different (x_1, x_2) coordinates in 4-D WDLs. Coordinates (128, 128), (64, 64), and (200, 200) have been selected. The results demonstrate the effectiveness of all three methods in capturing image features (Fig. 15). By selecting different coordinates, we can obtain diverse local features from the image. Looking ahead, these transformation methods can be further integrated with advanced algorithms like deep learning to enhance image analysis capabilities and enable more sophisticated applications. The combination of these analytical methods with deep learning holds great potential for unlocking new insights and advancements in the field of image analysis.

8. Conclusion

This article has presented a comprehensive discussion on three types of generalized two-dimensional (2-D) analytic signals (GASs), namely generalized 2-D complex analytic signal, generalized 2-D quaternionic analytic signal and generalized 2-D monogenic signal. These GASs have been defined based on the parametric Hilbert transform (PHT) and parametric Riesz transform (PRT) which are derived by the harmonic field and the linear canonical transform (LCT). The article has showcased the advantages of GASs in envelope detection, demonstrating their potential in practical applications. Moreover, the article has explored the four-dimensional Wigner distributions associated with the LCT (WDLs) for these GASs. The WDLs offer insights into the time-frequency characteristics of GASs and have shown promise for feature extraction purposes. The contributions of this article are significant, including the derivation of PHT and PRT, the definition of three distinct types of GASs, and the introduction of three types of WDLs based on correlation products. These contributions enhance our understanding of 2-D analytic signals and their applicability in various domains. In summary, this article provides valuable insights into the theoretical foundations and practical applications of 2-D analytic signals. It paves the way for further research and development in this field, offering po-

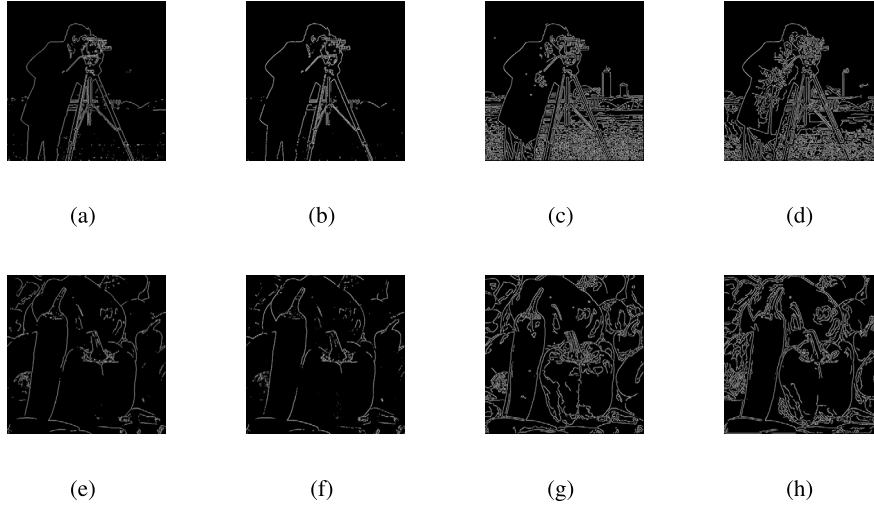


Fig. 12. Comparison of envelope detection effects using different methods (Cameraman image and Peppers image). (a) Sobel operator. (b) Roberts operator. (c) Canny operator. (d) Proposed method. (e) Sobel operator. (f) Roberts operator. (g) Canny operator. (h) Proposed method.

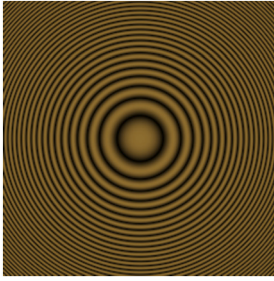


Fig. 13. The Newton's ring.

tential advancements in signal processing, image analysis, and other related areas.

CRediT authorship contribution statement

Jian-Yi Chen: original draft and conceptualization.
Bing-Zhao Li: review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Differential equation of 2-D harmonic function in linear canonical domain

Applying the 1-D LCT on (23) in the t_1 direction with $A = (a, b; c, d)$, we obtain

$$\begin{aligned} L_A \{ \text{div}(\nabla p(\mathbf{t})) \} &= L_A \left\{ \frac{\partial}{\partial t_1} e^{i \frac{a}{2b} t_1^2} v_1(\mathbf{t}) + \frac{\partial}{\partial t_2} e^{i \frac{a}{2b} t_1^2} v_2(\mathbf{t}) \right\} (\omega_1, t_2) = 0, \\ L_A \left\{ e^{-i \frac{a}{2b} t_1^2} \frac{\partial}{\partial t_1} e^{i \frac{a}{2b} t_1^2} v_1(\mathbf{t}) + e^{-i \frac{a}{2b} t_1^2} \frac{\partial}{\partial t_2} e^{i \frac{a}{2b} t_1^2} v_2(\mathbf{t}) \right\} (\omega_1, t_2) &= 0, \\ L_A \left\{ e^{-i \frac{a}{2b} t_1^2} \frac{\partial}{\partial t_1} e^{i \frac{a}{2b} t_1^2} v_1(\mathbf{t}) \right\} (\omega_1, t_2) + \frac{\partial}{\partial t_2} L_A \{ v_2 \} (\omega_1, t_2) &= 0, \end{aligned} \quad (\text{A.1})$$

where $\mathbf{t} = (t_1, t_2)^\top$. Based on the differential properties of the FT, we have

$$\begin{aligned} L_A \left\{ e^{-i \frac{a}{2b} t_1^2} \frac{\partial}{\partial t_1} e^{i \frac{a}{2b} t_1^2} v_1(\mathbf{t}) \right\} (\omega_1, t_2) &= \frac{1}{\sqrt{i2\pi b}} \int \frac{\partial}{\partial t_1} e^{i \frac{a}{2b} t_1^2} v_1(\mathbf{t}) e^{i(-\frac{1}{b} t_1 \omega_1 + \frac{d}{2b} \omega_1^2)} dt_1 \\ &= \frac{1}{\sqrt{i2\pi b}} e^{i \frac{d}{2b} \omega_1^2} \frac{\omega_1}{b} \int e^{i \frac{a}{2b} t_1^2} v_1(\mathbf{t}) e^{-i \frac{1}{b} t_1 \omega_1} dt_1 \\ &= i \frac{\omega_1}{b} L_A \{ v_1 \} (\omega_1, t_2). \end{aligned} \quad (\text{A.2})$$

Take (A.2) into (A.1), there is

$$i \frac{\omega_1}{b} L_A \{ v_1 \} (\omega_1, t_2) + \frac{\partial}{\partial t_2} L_A \{ v_2 \} (\omega_1, t_2) = 0. \quad (\text{A.3})$$

Since $v_k(\mathbf{t}) = e^{-i \frac{a}{2b} t_k^2} \frac{\partial}{\partial t_k} p(\mathbf{t})$, $k = 1, 2$, the equation (A.3) can be rewritten as

$$\begin{aligned} e^{i \frac{a}{2b} t_1^2} \frac{\omega_1}{b} L_A \left\{ e^{-i \frac{a}{2b} t_1^2} \frac{\partial}{\partial t_1} p(\mathbf{t}) \right\} (\omega_1, t_2) &+ e^{i \frac{a}{2b} t_1^2} \frac{\partial}{\partial t_2} L_A \left\{ e^{-i \frac{a}{2b} t_1^2} \frac{\partial}{\partial t_2} p(\mathbf{t}) \right\} (\omega_1, t_2) = 0, \\ i^2 \frac{\omega_1^2}{b^2} L_A \{ p \} (\omega_1, t_2) + \frac{\partial^2}{\partial t_2^2} L_A \{ p \} (\omega_1, t_2) &= 0, \\ - \frac{\omega_1^2}{b^2} L_A \{ p \} (\omega_1, t_2) + \frac{\partial^2}{\partial t_2^2} L_A \{ p \} (\omega_1, t_2) &= 0. \end{aligned} \quad (\text{A.4})$$

Therefore, the differential equation of the harmonic potential $p(\mathbf{t})$ in the linear regular domain is derived.

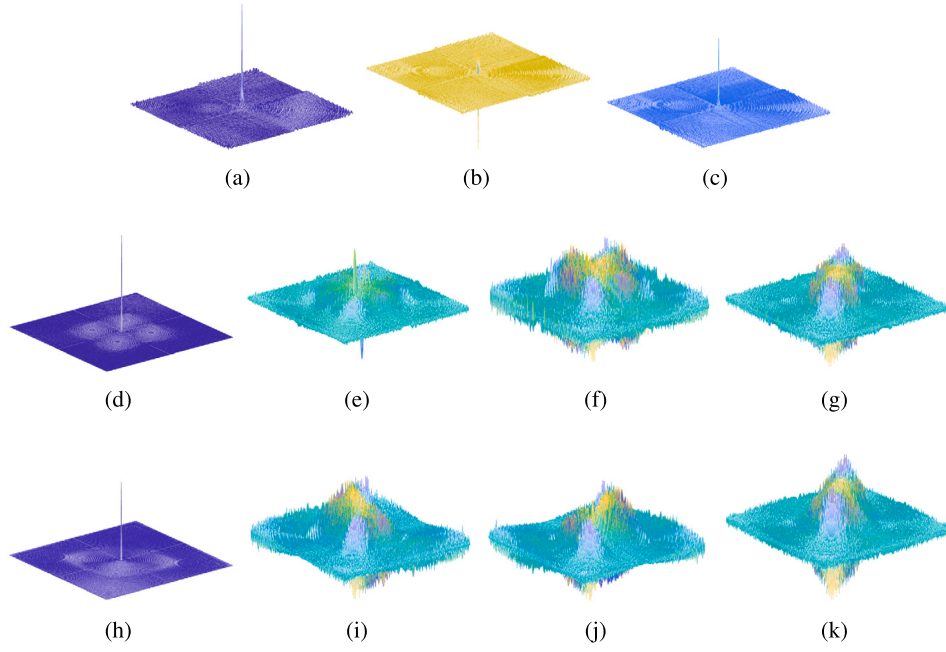


Fig. 14. The 4-D WDLs for the GCS, QGS, and GMS of the Newton's ring in 13 with $A_1 = A_2(0, 1/2, -2, 0)$, $A_3 = A_4 = (0, 0.9999, -1/0.9999, 0)$. (a-c) The modulus, real-part and imaginary-part of $W_{\hat{\Lambda},1}(128, 128, \omega_1, \omega_2)$. (d-g) The modulus, i-part, j-part and k-part of $W_{\hat{\Lambda},Q}(128, 128, \omega_1, \omega_2)$. (h-k) The modulus, i-part, j-part and k-part of $W_{\hat{\Lambda},M}(128, 128, \omega_1, \omega_2)$.

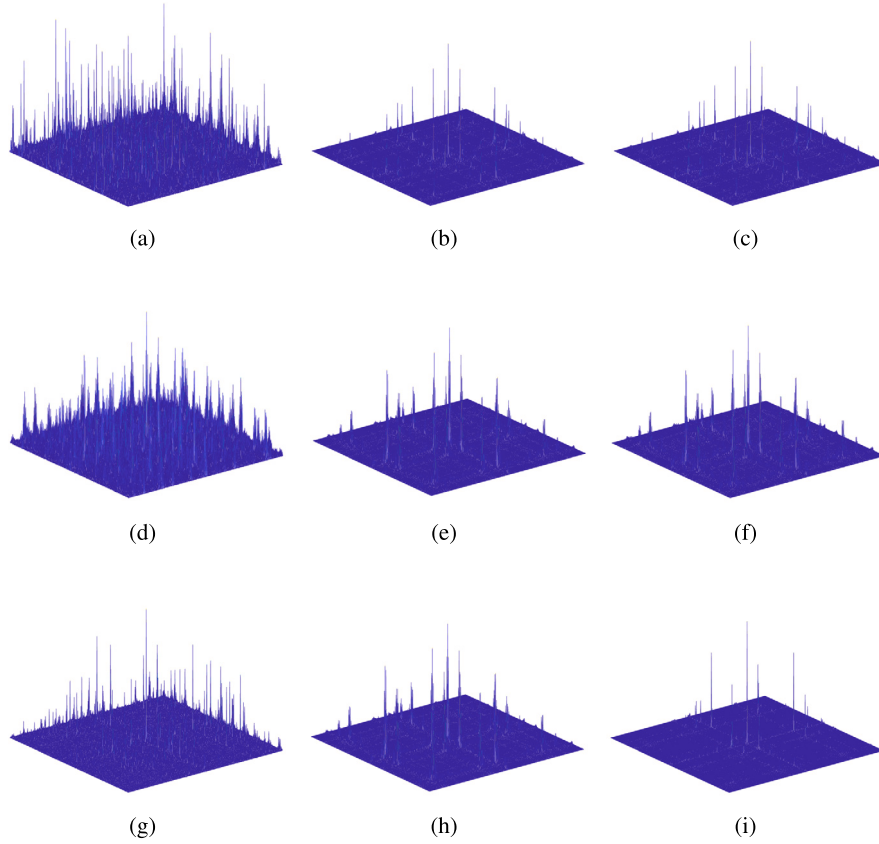


Fig. 15. The modulus of the 4-D WDLs for the GCS, QGS, and GMS of Lena image with $A_1 = (100, 9, -1/9, 0)$, $A_2 = (100, 9, -1/9, 0)$, $A_3 = (3, 0.9, 1/0.9, 0)$, $A_4 = (3, 0.9, 1/0.9, 0)$. (a) $W_{\hat{\Lambda},1}(128, 128, \omega_1, \omega_2)$. (b) $W_{\hat{\Lambda},Q}(128, 128, \omega_1, \omega_2)$. (c) $W_{\hat{\Lambda},M}(128, 128, \omega_1, \omega_2)$. (d) $W_{\hat{\Lambda},1}(64, 64, \omega_1, \omega_2)$. (e) $W_{\hat{\Lambda},Q}(64, 64, \omega_1, \omega_2)$. (f) $W_{\hat{\Lambda},M}(64, 64, \omega_1, \omega_2)$. (g) $W_{\hat{\Lambda},1}(200, 200, \omega_1, \omega_2)$. (h) $W_{\hat{\Lambda},Q}(200, 200, \omega_1, \omega_2)$. (i) $W_{\hat{\Lambda},M}(200, 200, \omega_1, \omega_2)$.

Appendix B. Expanded forms of WDLs and AFLs of GQS and GMS

Let $\alpha_i = -\left(\frac{a_i}{2b_i}\tau_i^2 - \frac{1}{b_i}\tau_i\omega_i + \frac{d_i}{2b_i}\omega_i^2\right)$, $r_i = \sqrt{4\pi^2 b_i^2}$, $\theta_i = \arctan(2\pi b_i)$, $i = 1, 2$, then the insertion of (52) in (54) yields the following form of $W_{A,Q}$

$$\begin{aligned} W_{A,Q}(t_1, t_2, \omega_1, \omega_2) &= \iint \frac{1}{\sqrt{r_1}} e^{-i\left(\frac{\theta_1}{2} + \alpha_1\right)} [A_Q + iB_Q + jC_Q + kD_Q] \frac{1}{\sqrt{r_2}} e^{-i\left(\frac{\theta_2}{2} + \alpha_2\right)} d\tau_1 d\tau_2 \\ &= \frac{1}{\sqrt{r_1 r_2}} \iint [A_Q + iB_Q + jC_Q + kD_Q] \cos\left(\frac{\theta_1}{2} + \alpha_1\right) d\tau_1 \cos\left(\frac{\theta_2}{2} + \alpha_2\right) d\tau_2 \\ &\quad + \frac{1}{\sqrt{r_1 r_2}} \iint [-iA_Q + B_Q - kC_Q + jD_Q] \sin\left(\frac{\theta_1}{2} + \alpha_1\right) d\tau_1 \cos\left(\frac{\theta_2}{2} + \alpha_2\right) d\tau_2 \\ &\quad + \frac{1}{\sqrt{r_1 r_2}} \iint [-jA_Q - kB_Q + C_Q + iD_Q] \cos\left(\frac{\theta_1}{2} + \alpha_1\right) d\tau_1 \sin\left(\frac{\theta_2}{2} + \alpha_2\right) d\tau_2 \\ &\quad + \frac{1}{\sqrt{r_1 r_2}} \iint [kA_Q - jB_Q - iC_Q + D_Q] \cos\left(\frac{\theta_1}{2} + \alpha_1\right) d\tau_1 \sin\left(\frac{\theta_2}{2} + \alpha_2\right) d\tau_2. \end{aligned} \quad (B.5)$$

Then reorder the terms and apply a shorter notation of integrals $\int \{(\cdot)d\tau_1\}(\cdot)d\tau_2$, which yields

$$\begin{aligned} W_{A,Q}(t_1, t_2, \omega_1, \omega_2) &= W_{A,Qr} + iW_{A,Qi} + jW_{A,Qj} + kW_{A,Qk} \\ &= \frac{1}{\sqrt{r_1 r_2}} \iint [A_Q \cos\left(\frac{\theta_1}{2} + \alpha_1\right) \cos\left(\frac{\theta_2}{2} + \alpha_2\right) \\ &\quad + B_Q \sin\left(\frac{\theta_1}{2} + \alpha_1\right) \cos\left(\frac{\theta_2}{2} + \alpha_2\right) \\ &\quad + C_Q \cos\left(\frac{\theta_1}{2} + \alpha_1\right) \sin\left(\frac{\theta_2}{2} + \alpha_2\right) \\ &\quad + D_Q \sin\left(\frac{\theta_1}{2} + \alpha_1\right) \sin\left(\frac{\theta_2}{2} + \alpha_2\right)] d\tau_1 d\tau_2 \\ &\quad + \frac{i}{\sqrt{r_1 r_2}} \iint [B_Q \cos\left(\frac{\theta_1}{2} + \alpha_1\right) \cos\left(\frac{\theta_2}{2} + \alpha_2\right) \\ &\quad - A_Q \sin\left(\frac{\theta_1}{2} + \alpha_1\right) \cos\left(\frac{\theta_2}{2} + \alpha_2\right) \\ &\quad + D_Q \cos\left(\frac{\theta_1}{2} + \alpha_1\right) \sin\left(\frac{\theta_2}{2} + \alpha_2\right) \\ &\quad - C_Q \sin\left(\frac{\theta_1}{2} + \alpha_1\right) \sin\left(\frac{\theta_2}{2} + \alpha_2\right)] d\tau_1 d\tau_2 \\ &\quad + \frac{j}{\sqrt{r_1 r_2}} \iint [C_Q \cos\left(\frac{\theta_1}{2} + \alpha_1\right) \cos\left(\frac{\theta_2}{2} + \alpha_2\right) \\ &\quad + D_Q \sin\left(\frac{\theta_1}{2} + \alpha_1\right) \cos\left(\frac{\theta_2}{2} + \alpha_2\right) \\ &\quad - A_Q \cos\left(\frac{\theta_1}{2} + \alpha_1\right) \sin\left(\frac{\theta_2}{2} + \alpha_2\right) \\ &\quad - B_Q \sin\left(\frac{\theta_1}{2} + \alpha_1\right) \sin\left(\frac{\theta_2}{2} + \alpha_2\right)] d\tau_1 d\tau_2 \\ &\quad + \frac{k}{\sqrt{r_1 r_2}} \iint [D_Q \cos\left(\frac{\theta_1}{2} + \alpha_1\right) \cos\left(\frac{\theta_2}{2} + \alpha_2\right) \\ &\quad - C_Q \sin\left(\frac{\theta_1}{2} + \alpha_1\right) \cos\left(\frac{\theta_2}{2} + \alpha_2\right) \\ &\quad - B_Q \cos\left(\frac{\theta_1}{2} + \alpha_1\right) \sin\left(\frac{\theta_2}{2} + \alpha_2\right) \\ &\quad + A_Q \sin\left(\frac{\theta_1}{2} + \alpha_1\right) \sin\left(\frac{\theta_2}{2} + \alpha_2\right)] d\tau_1 d\tau_2. \end{aligned} \quad (B.6)$$

Similarly, let $\beta_i = -\left(\frac{a_i}{2b_i}t_i^2 - \frac{1}{b_i}t_i\omega_i + \frac{d_i}{2b_i}\omega_i^2\right)$, then the insertion of (52) in (55) yields the following form of $AF_{A,Q}$

$$\begin{aligned} AF_{A,Q}(\tau_1, \tau_2, \omega_1, \omega_2) &= \iint \frac{1}{\sqrt{r_1}} e^{-i\left(\frac{\theta_1}{2} + \beta_1\right)} [A_Q + iB_Q + jC_Q + kD_Q] \frac{1}{\sqrt{r_2}} e^{-i\left(\frac{\theta_2}{2} + \beta_2\right)} d\tau_1 d\tau_2 \\ &= AF_{A,Qr} + iAF_{A,Qi} + jAF_{A,Qj} + kAF_{A,Qk}. \end{aligned} \quad (B.7)$$

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